PHYSICS BOARD 2026: THE ULTIMATE STRATEGIC ROADMAP

Syllabus-Optimized Analysis of 2023-2025 Board Papers

CHAPTER-WISE WEIGHTAGE & TRENDS

Chapter	Importance	Annual Trend
Ray Optics & Instruments	🚨 VERY HIGH	Telescope/Microscope Diagrams + Lens Maker
Semiconductors	🚨 VERY HIGH	Full Wave Rectifiers & Energy Bands
Current Electricity	⚡ HIGH	Drift Velocity & Internal Resistance
Atoms & Nuclei	⚡ HIGH	Bohr Radius Derivation & Nuclear Density
Electrostatics	⚡ HIGH	Capacitors with Dielectrics & Dipoles

MOST REPEATED TOP 10 TOPICS

1. **Rectifiers (Full Wave):** Diagram + Waveforms.
2. **Optical Instruments:** Astronomical Telescope / Compound Microscope diagrams.
3. **Bohr Model:** Proving $r \propto n^2$ and Energy E derivation.
4. **Nuclear Density:** Constant ratio (1:1) conceptual/MCQ.
5. **Drift Velocity:** Relation with current ($I = nAev_d$).
6. **Capacitance:** Effect of Dielectric slab ($t < d$).
7. **Moving Coil Galvanometer:** Radial field and conversion to Ammeter.
8. **Huygens' Principle:** Proof of laws of refraction.
9. **Force b/w Parallel Wires:** Ampere's definition.
10. **Photoelectric Effect:** Stopping potential vs Frequency graphs.

PREDICTION ZONE (2026 BOARD)

🔥 **MUST PREPARE:** Optical Instruments at infinity/normal adjustment. This is the most consistent 5-marker.

⚡ **HIGH CHANCE:** Lens Maker's Formula derivation and Full Wave Rectifier working.

EXPECTED PRACTICE SET (MUST-SOLVE)

1. [MCQ] Two nuclei have mass numbers 1:125. Their nuclear density ratio is: **Ans: 1:1 (Independent of A)**

2. [MCQ] In forward bias of a P-N junction, the depletion layer: **Ans: Decreases**

3. [Derivation] Derive the expression for the radius of n^{th} orbit in Bohr's model. Prove $r \propto n^2$.

4. [Diagram] Draw a full wave rectifier circuit. Show input and output waveforms.

5. [Concept] Which part of EM spectrum is used in water purifiers? **Ans: UV Rays**

LAST 7 DAYS QUICK REVISION & TRAPS

⚠️ **COMMON STUDENT TRAPS:**

- Confusing **Interference** (equal width) with **Diffraction** (central max double width).
- Thinking Nuclear Density changes with Mass—it is **Constant**.
- Forgetting to convert `cm` to `m` in Ray Optics numericals.
- Writing magnetic flux unit as Tesla (It's **Weber**; Tesla is for Field).

Formula Cheat-Sheet:

Lens Maker: $1/f = (n-1)(1/R_1 - 1/R_2)$	Fringe Width: $ eta = D\lambda/d$
Drift Velocity: $v_d = eE au/m$	Photoelectric: $K_{max} = h u - \phi_0$